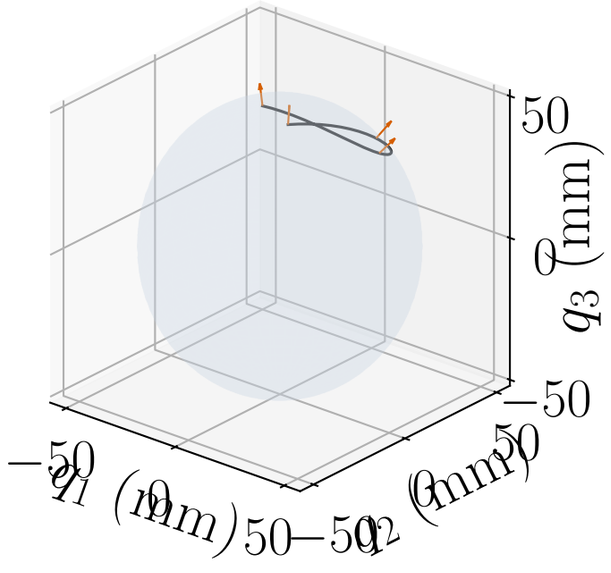
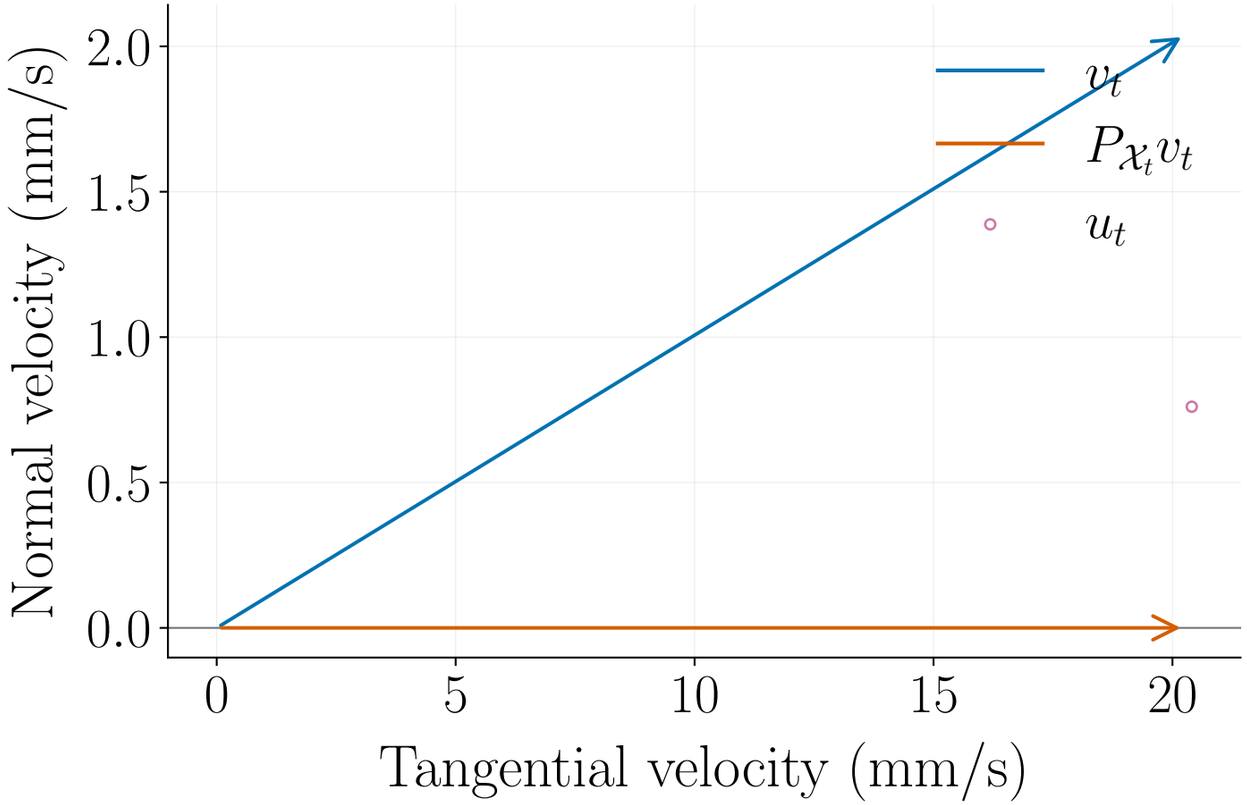

50
0
−50
q_3 (mm)
−50
0
50
q_2 (mm)
−50
0
50
q_1 (mm)


Normal velocity (mm/s)
2.0
1.5
1.0
0.5
0.0
0
5
10
15
20
Tangential velocity (mm/s)
v_t
$P_{\mathcal{X}_t} v_t$
u_t